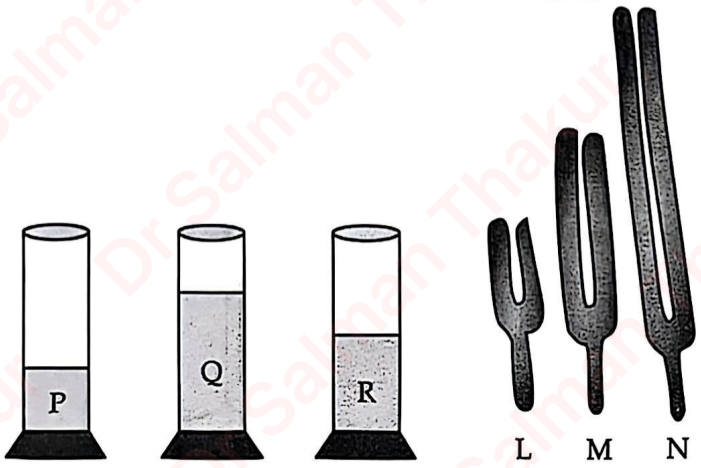
Chapter 5 Sound

1 Objective Type Questions 1 mark each

A Multiple Choice Questions

1. The below picture shows three cylinders filled with water to different extents. The tuning forks' L, M and N in vibration produce louder sound when held above the mouth of the cylinders. Which combination is correct for hearing a louder sound of the tuning fork?

A & Ap [CFPQ] M

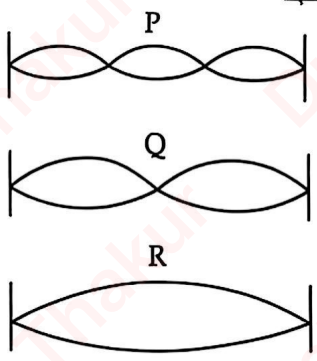


(a) L–R, M–Q, N–P
(b) L–P, M–Q, N–R
(c) L–R, M–P, N–Q
(d) L–Q, M–R, N–P

Concept to be Applied: Resonance

2. What is the ratio of wavelengths between P, Q and R?

AI A & Ap [CFPQ] M



(a) 3:2:1
(b) 1:2:3
(c) 6:3:2
(d) 2:3:6

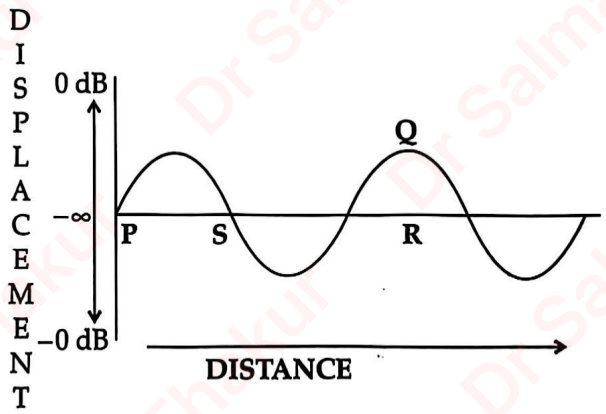
3. During a science experiment, a student uses a tuning fork with a frequency of 512 Hz. She strikes the tuning fork in air and water and records the time for sound to travel 1 m in both media.

If the speed of sound in air is 340 m/s and in water is 1480 m/s, how much time will the sound take to travel 1 m in water? AI Ap M

(a) 0.00294 s
(b) 0.000676 s
(c) 0.67 s
(d) 1.48 s

Concept to be Applied: Speed of sound in different media

4. Sumit, a teenager, enjoys playing his music loudly. However, his grandmother consistently lowers the volume of the music player. Analyse the wave to identify which characteristic of the wave has been altered when the grandmother reduces the volume. A [CFPQ] M



(a) The length of PS becomes more.
(b) The length of SR becomes less.
(c) The length of QR becomes less
(d) The length of PR becomes less.

Concept to be Applied: Loudness

5. A storm is approaching a village and the villagers see a flash of lightning. A few seconds later, they hear the thunder. A student observing this wants to understand why the sound arrives later than the light.

Why do the villagers hear thunder after seeing the lightning? U E

(a) Light travels slower than sound.
(b) Sound requires a medium, while light does not.
(c) Light travels faster than sound in air.
(d) Lightning causes sound to travel slower.

Concept to be Applied: Propagation of sound and light in different media

B Match the Columns

Directions: Match Column I with Column II and choose the correct option.

1. Match the following: U M

Column I	Column II
(A) Infrasonic waves	(i) Below 20 Hz
(B) Sonic waves	(ii) 20 Hz to 20,000 Hz
(C) Ultrasonic waves	(iii) Above 20,000 Hz
(D) Supersonic speed	(iv) Speed greater than the speed of sound

(a) A (ii), B (iii), C (iv), D (i)
(b) A (i), B (ii), C (iii), D (iv)
(c) A (iv), B (iii), C (ii), D (i)
(d) A (ii), B (iv), C (iii), D (i)

Concept to be Applied: Frequency ranges (Infrasonic, Sonic, Ultrasonic)

2. Match the following: Ap E

Column I	Column II
(A) Air	(i) Sound cannot travel
(B) Water	(ii) 5,960 m/s
(C) Steel	(iii) 1,480 m/s
(D) Vacuum	(iv) 343 m/s

(a) A (ii), B (iii), C (i), D (iv)
(b) A (i), B (ii), C (iii), D (iv)
(c) A (iv), B (iii), C (ii), D (i)
(d) A (iii), B (i), C (iv), D (ii)

Concept to be Applied: Speed of sound in different media

3. Match the following: R E

Column I	Column II
(A) v	(i) Frequency of sound
(B) f	(ii) Velocity of sound
(C) λ	(iii) Time period
(D) T	(iv) Wavelength of sound

(a) A (ii), B (i), C (iv), D (iii)
(b) A (i), B (ii), C (iii), D (iv)
(c) A (iii), B (ii), C (i), D (iv)
(d) A (iv), B (iii), C (i), D (ii)

4. Match the following: AI U M

Column I	Column II
(A) Speed of sound	(i) Much slower than the speed of light
(B) Speed of light	(ii) Approximately 300,000,000 m/s
(C) Thunder	(iii) Heard after lightning due to slower speed of sound
(D) Lightning	(iv) Seen before thunder due to faster speed of light

(a) A (iv), B (ii), C (iii), D (i)
(b) A (iv), B (iii), C (ii), D (i)
(c) A (ii), B (iii), C (i), D (iv)
(d) A (i), B (ii), C (iii), D (iv)

Concept to be Applied: Propagation of sound and comparison with light speed

5. Match the following: AI A M

Column I	Column II
(A) Higher temperature	(i) Increases the speed of sound slightly
(B) Denser medium	(ii) No sound propagation
(C) Higher humidity	(iii) Increases speed of sound
(D) Vacuum	(iv) More speed of sound

(a) A (i), B (ii), C (iii), D (iv)
(b) A (iii), B (iv), C (i), D (ii)
(c) A (ii), B (iii), C (iv), D (i)
(d) A (iv), B (iii), C (ii), D (i)

Concept to be Applied: Factors affecting the speed of sound

C Fill in the Blanks

Direction: Choose the correct answer among the given options (a), (b), (c) and (d).

1. _____ of UV > _____ of microwaves
The quantity suitable for filling in both the blanks is. R [CFPQ] E

(a) speed in vacuum
(b) speed in medium
(c) frequency
(d) wavelength

Concept to be Applied: Electromagnetic spectrum

2. In a thunderstorm, a student hears the sound of thunder 5 s after seeing a flash of lightning. The speed of sound in air is 340 m/s, while the speed of light is approximately 3×10^8 m/s.
The distance between the student and the lightning strike is approximately _____ m. AI Ap M

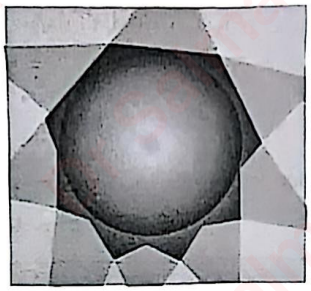
(a) 680 **(b)** 1700
(c) 3400 **(d)** 5000

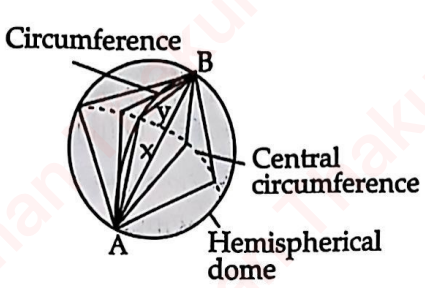
D Assertion—Reason Based Questions

Directions: In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

1. The diagram shows the magnificent dome at Gol Ghumbaz at Bijapur. A boy standing at A whispers, and it is heard at B across the huge dome. [Speed of sound is 330 ms^{-1}]





Assertion: The distance travelled by sound waves $(x + y)$ should be < 33 m.
Reason: Sound reflected within 0.1 s is not heard by humans. A & Ap [CFPQ] Ⓜ

2. Assertion: Ultrasonic waves can be used in medical imaging.
Reason: Ultrasonic waves have higher frequencies than audible sounds. AI Ap Ⓜ
Concept to be Applied: Ultrasonic wave applications

3. Assertion: The speed of sound increases with an increase in temperature.
Reason: Higher temperatures cause particles in a medium to move more rapidly, facilitating faster sound wave propagation. Ap Ⓜ
Concept to be Applied: Factors affecting the speed of sound

4. Assertion: Sound waves travel faster in air than in water.
Reason: Air is less dense than water. AI U Ⓜ
Concept to be Applied: Speed of sound in different media

5. Assertion: Infrasonic waves can be detected by the human ear.
Reason: Infrasonic waves have frequencies below 20 Hz. U Ⓔ
Concept to be Applied: Properties of Infrasonic sound

E Statement-Based Questions

Directions: In the following questions, two statements are given.

Mark the correct choice as:
(a) Statement I is incorrect, but statement II is correct.
(b) Both statement I and statement II are correct.
(c) Both statement I and statement II are incorrect.
(d) Statement I is correct, but statement II is incorrect.

1. Consider the following statements:
Statement I: Sound travels faster in water than in air.
Statement II: Water is denser than air. U Ⓜ
Concept to be Applied: Speed of sound in different media

2. Consider the following statements:
Statement I: Sound travels slower in solids than in gases.
Statement II: Solids are denser than gases. AI U Ⓜ

3. Consider the following statements:
Statement I: Thunder is heard before lightning is seen during a thunderstorm.
Statement II: Sound travels faster than light. U Ⓔ
Concept to be Applied: Speed of sound vs light

2 Subjective Type Questions

A VSATQs

2 marks each

1. John is trying to create musical notes using water-filled bottles, as shown in the picture.



Which of the bottles is likely to produce a musical note with the highest pitch? Give reasons. A [CFPQ] Ⓜ
Concept to be Applied: Pitch of Sound

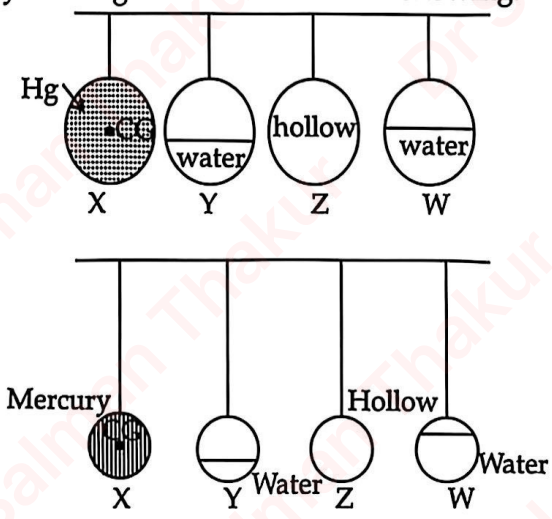
2. When Shyam, the band leader, struck both the bass drum and the kettle drum with equal force, the bass drum emitted a sound measuring 120 dB, whereas the kettle drum produced a sound of 90 dB. Explain this discrepancy in the loudness. Ap [CFPQ] Ⓜ
Concept to be Applied: Sound Intensity and Decibel Levels.

3. Ashish, Sameer, Aditya and Mohit were sitting on the terrace of Ashish's house listening to the songs playing on Aditya's mobile phone. Mohit, sitting in one corner, a little distance away from the others, requests to increase the volume of the songs. But Aditya says the volume is full. Sameer, being a smart guy, gets up, goes into the kitchen, brings an empty steel glass, places the mobile phone inside it, and solves the problem. What is the reason for Sameer to keep the mobile phone in the glass?
Ap [CFPQ] Ⓜ
Concept to be Applied: Resonance and sound amplification

B SATQs

3 marks each

1. Study the diagram and answer the following:



(a) Arrange the following pendulums according to their natural frequencies (use $<$ $>$ $=$).
(b) If Pendulum Z is initiated into oscillation, which pendulum among the others will exhibit the highest amplitude of vibration?

(c) Give reasons for your answer. A [CFPQ] Ⓜ

Concept to be Applied: Resonance and sound amplification

2. Seismic waves have different frequencies. During earthquakes, why are short-length buildings more prone to damage caused by high-frequency seismic waves?

U [CFPQ] Ⓜ

Concept to be Applied: Resonance and natural frequency of oscillation.

3. A student in a laboratory taps a metal rod and notices that the sound is transmitted clearly through the rod but is barely heard when tapping the rod in the air.

Why does the sound travel more efficiently through the metal rod than through the air? U Ⓜ

4. During a thunderstorm, a student notices a time gap between seeing lightning and hearing thunder.

Explain why the student hears thunder after seeing the lightning. AI Ap Ⓔ

Concept to be Applied: Comparison of the speed of sound with the speed of light

5. A factory uses ultrasonic waves for cleaning delicate instruments.

Why are ultrasonic waves used for cleaning, and not infrasonic or sonic waves? AI Ap Ⓜ

Concept to be Applied: Infrasonic, sonic, ultrasonic frequencies and their applications

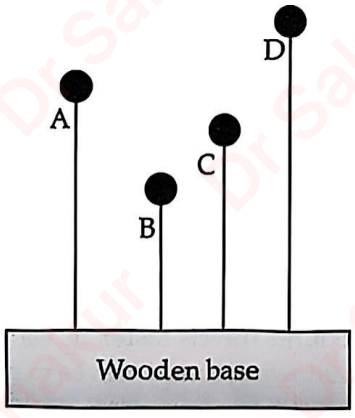
6. A musician plays a guitar and the pitch of the sound changes as they tighten a string.

How does tightening the guitar string affect the frequency and pitch of the sound? A Ⓜ

C LATQs

4 marks each

1. The adjacent diagram shows four solid plastic balls with wires fitted on a wooden base. A person shakes the wooden base to and fro (forward and backward) periodically. It is observed that even the balls start vibrating. It is also observed that all balls vibrate, but only one ball vibrates vigorously.



(a) Explain why only one ball vibrates vigorously.

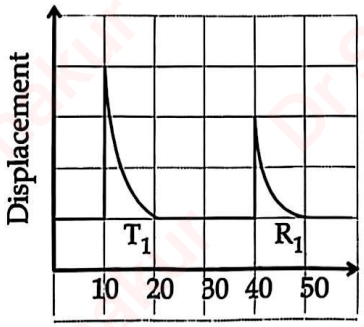
(b) If f_A, f_B, f_C and f_D are the natural frequencies of vibration of the wires, then arrange them in the increasing order of their frequencies and give reasons for the same. A [CFPQ] Ⓜ

2. Examine the graph depicting the transmitted (T) and reflected (R) waves from submarines A and B, which are equipped with a device emitting ultrasonic sounds. Study the graph and answer the following. (Speed of sound in water is 1500 m/s)

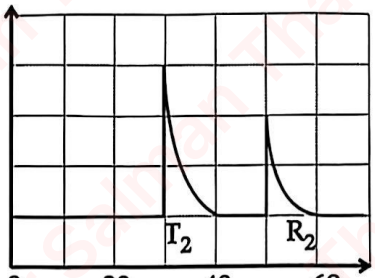
(a) Name the device used here.

(b) Why is ultrasonic sound used in this device?

(c) Which submarine is closer to the ship? Calculate the distance of the closest submarine. R & E [CFPQ] Ⓓ



submarine A



submarine B

3. During a thunderstorm, you notice that you see the lightning almost instantly but hear the thunder after a few seconds. You also observe that on different occasions, the delay between the lightning flash and thunder sound varies.

(a) Explain the reason for this difference in the perception of lightning and thunder.

(b) Calculate the time delay between the lightning flash and thunder if the distance of the storm is 2 km and the speed of sound is 340 m/s. U Ⓜ

Concept to be Applied: Comparison of speed of sound and light, sound propagation, $v = f\lambda$

4. A doctor is performing an ultrasound scan to examine the internal organs of a patient. The machine emits high-frequency sound waves that are beyond the range of human hearing. The doctor explains that these waves help in producing images of the body's internal structure.

(a) Explain the working principle of ultrasound and distinguish between ultrasonic and supersonic waves.

(b) What are the medical applications of ultrasonic waves? Ap Ⓜ

Concept to be Applied: Ultrasonic and supersonic waves, application of ultrasonic waves